

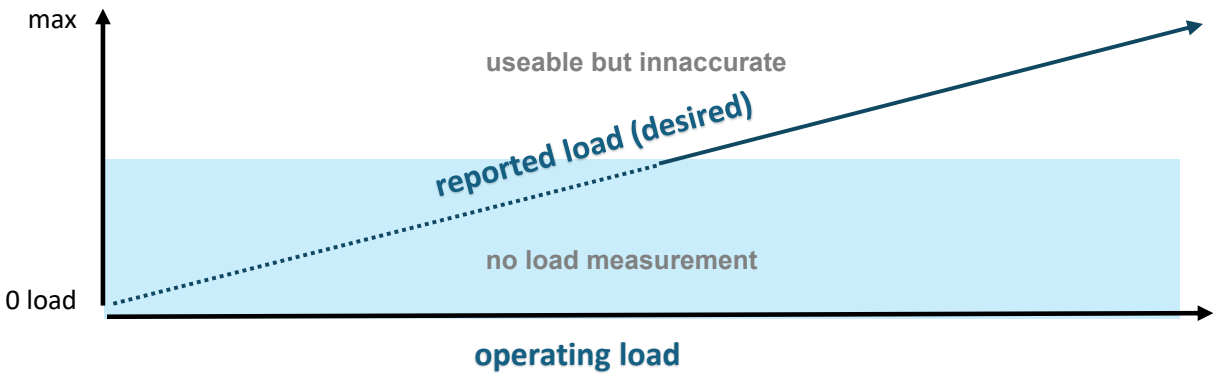
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9A Press Load Monitoring

ULTIMATE GOAL

Current situation:

The load monitoring system on 9A press is not accurate.
 The system cannot form a measurement at <50% of press load capacity.
 The measurement is based on pressure readings from hydraulic overload protector system.



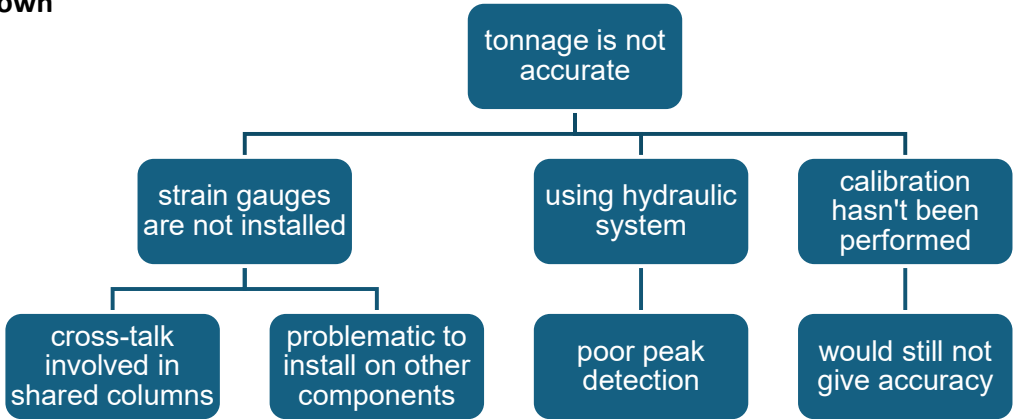
Ideal Situation:

Load monitoring system takes accurate measurement through entire range
 the system does not require frequent maintenance

Gap Identified:

accuracy needs improvement

Problem Break Down



Target

- ▶ install/calibrate a system to read out peak tonnage
- ▶ use system on press lines with column-sharing slides
- ▶ Tonnage reads a number that matches same dies tonnage on other machines (accurate)
- ▶ Tonnage reads consistent numbers (precise)

Root Cause Analysis

no strain gauge system installed on presses
 strain gauges read cross talk within columns
 logic does not compensate for 2 slides
 does not gauge which side the load comes from
 only one strain gauge on each column
 system is not calibrated for shared columns

Countermeasures

	accurate	precise	cost	maintenance	simple install	OVERALL
Custom strain gauge system	O	O	Δ	Δ	O	O
Calibrate the current system	X	Δ	Δ	Δ	Δ	Δ
Do nothing	X	X	O	O	Δ	Δ
Install strain gauge on slides	Δ	Δ	X	Δ	Δ	Δ

Schedule/Assignments

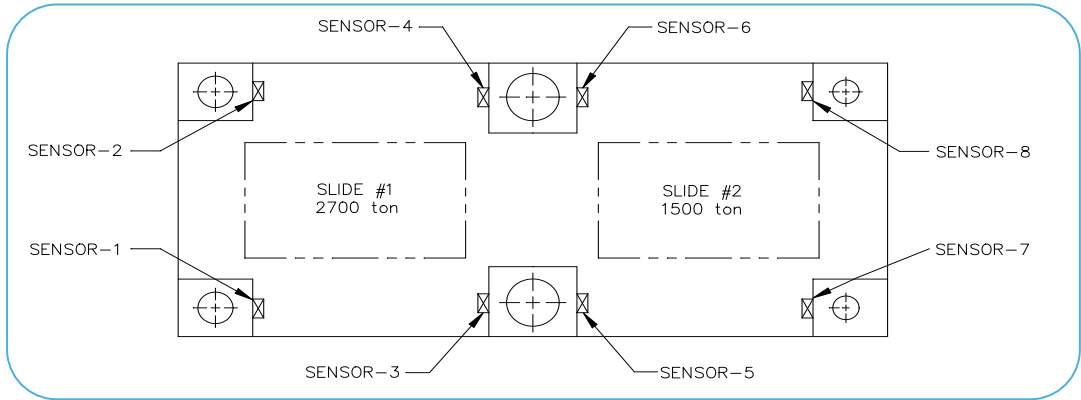
What	Who	Time schedule:
planning	Toyota PE	?
install	Contractor	?
calibration	Both	?

Details:

- Install strain gauge system for shared slides and use special calibration
- ▶ Gauges will measure left and right sides of shared columns
 - ▶ Calibrate gauges simultaneously with load-cell dies
 - ▶ Install Allen-Bradley HMI to readout peaks
 - ▶ Cost is \$(?????) for material and contractor installation.

Example of installation on shared columns

STAMPING PRESS PLAN VIEW



Compiled Notes for A3 Problem Solving on Stamping Press

Notes from my internship in 2009 in kentucky, having 3-4 months experience in stamping department, as an exercise in Toyota's Problem Solving Methods Reformatted in 2026.05 for readability & sharing online. This is the engineer's first attempt at an A3, with 2.5 years of study in Mech Eng. degree.

Problem Identification

Problem

Modular transfer press load monitor is inaccurate

OCY14Z

Target setting

Targets

- ▶ useable below 50% of press tonnage
- ▶ accurate tonnage reading
- ▶ not expensive
- ▶ little maintenance required

Brainstorm Ideas

Pin cushion pressure gauge monitors pressure on pin cushion accurately.

- ▶ the pressure gauge for the overload system is already very appropriate for it's application.
- ▶ the problem does not lie in the sensor but in the hydraulic system- it is not accurate enough

Strain gauges could be added to the slide

- ▶ attachment (welding, tapping, etc) could possibly damage plungers
- ▶ wires from strain gauges will be easily damaged.
radio transmitter could transmit signal from atop slide to eliminate wires

Strain gauges could be attached near bearing mounts on shaft?

Strain gauges mounted on the tie rods of each column

- ▶ can be calibrated for shared columns?
- ▶ two tie rods per column in shared columns?

Actions to Take

My notes did not reach a stage to establish actions but still included the anticipated parties to take action.

What	Who	Time schedule:
planning	Toyota PE	?
install	Contractor	?
calibration	Both	?